

Part 1: Text Processing and Exploratory Data Analysis

1.1. Data preparation

1. As a first step, you must pre-process the documents. In particular, for the text fields (title, description) you should:

- **Removing stop words**
- **Tokenization**
- **Removing punctuation marks**
- **Stemming**
- **and... anything else you think it's needed (bonus point)**

We have created a function that receives a line and does all the preprocess steps required plus transforming to lowercase. This will help us in the future when ranking the searches, because we will avoid that the same word written in lowercase or uppercase gets different results. As part of the data processing, we have converted to int the columns `selling_price` and `actual_price`, which were strings. We have also converted the rating to float. And finally, we have eliminated the "% off" from the discount, in order to leave only the value of discount to help us in the order by discount.

2. Take into account that for future queries, the final output must return (when present) the following information for each of the selected documents: pid, title, description, brand, category, sub_category, product_details, seller, out_of_stock, selling_price, discount, actual_price, average_rating, url

To achieve this, we have defined a list of the features we want to show when we ask for the information of a product. Then, we only show the selected columns when showing the dataset of products.

3. Decide how to handle the fields category, sub_category, brand, product_details, and seller during pre-processing. Should they be merged into a single text field, indexed as separate fields in the inverted index or any other alternative? Justify your choice, considering how their distinctiveness may affect retrieval effectiveness. What are the pros and cons of each approach?

Merged in a single text field:

- Pros:
 - It consumes less memory
 - It is simpler to implement
 - Less execution time
 - More precise search, if an item does not match the query closely, it will not be retrieved.
- Cons:
 - We have less control over the search
 - Not all fields have the same importance
 - There is a loss of field structure (a word is treated the same in brand or seller)

Separate fields:

- Pros:
 - More control over the search, we can boost important fields
 - It allows filtering for more specific searches
 - Queries are easier to interpret
- Cons:
 - The setup is more complex
 - It consumes more memory
 - Higher execution time

Our idea is to combine category and sub_category into a single text field, as this approach saves space and allows users to find similar items within the main category when searching for a specific piece.

On the other hand, we believe it is better to keep brand, seller, and product_details as separate fields in the inverted index. This helps retrieve products that better match users' preferences and keywords. For instance, if someone searches for "Nike", the system should clearly prioritize and filter products from that brand.

We have used AI tools to help us consider if our approach was logical or if it was a bad idea.

4. Consider the fields out_of_stock, selling_price, discount, actual_price, and average_rating. Decide how these should be handled during pre-processing to use in further search. Should they be indexed as textual terms?

These fields should not be indexed as textual terms, since they do not carry semantic or descriptive information that contributes to retrieval. Instead, they represent numeric or boolean attributes that describe quantitative aspects of a product (such as price, availability, or rating).

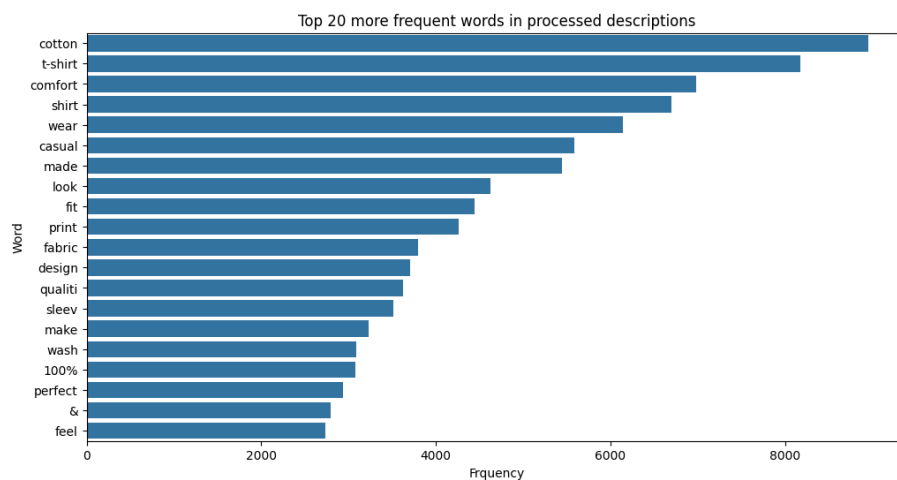
Therefore, these attributes should be handled as numeric or boolean fields rather than analyzed text. This allows them to be used effectively for filtering and ranking operations, such as filtering out products that are out of stock, sorting by lowest price, or boosting items with higher discounts or better ratings. Treating them as numeric attributes also ensures more efficient indexing and enables accurate range queries and scoring functions, which would not be possible if they were processed as textual data.

1.2. Exploratory Data Analysis

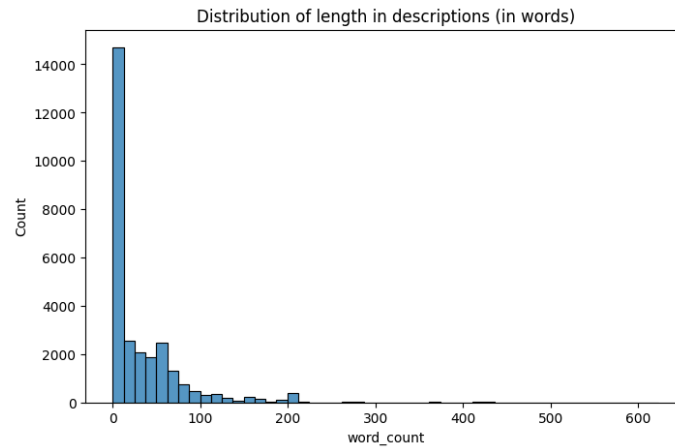
We have obtained the following plots:

	selling_price	discount	actual_price
count	28078.000000	27225.000000	27303.000000
mean	705.635088	50.256896	1455.528110
std	549.681489	16.887287	939.977456
min	99.000000	1.000000	150.000000
25%	390.000000	40.000000	849.000000
50%	545.000000	53.000000	1199.000000
75%	820.000000	63.000000	1799.000000
max	7999.000000	87.000000	12999.000000

In this plot we see the main statistics for each of the numerical variables, which are `selling_price`, `discount`, and `actual_price`.



In this plot we see the more frequent words. Mainly, they are related to the material, the type of cloth and the model.



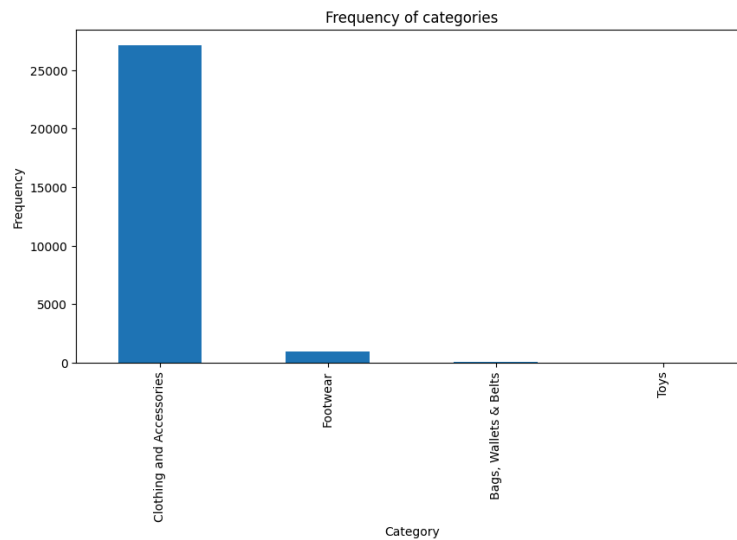
Here we see that in this context, the sentences tend to be short, but there are some cases where they can be up to 100 or more words.



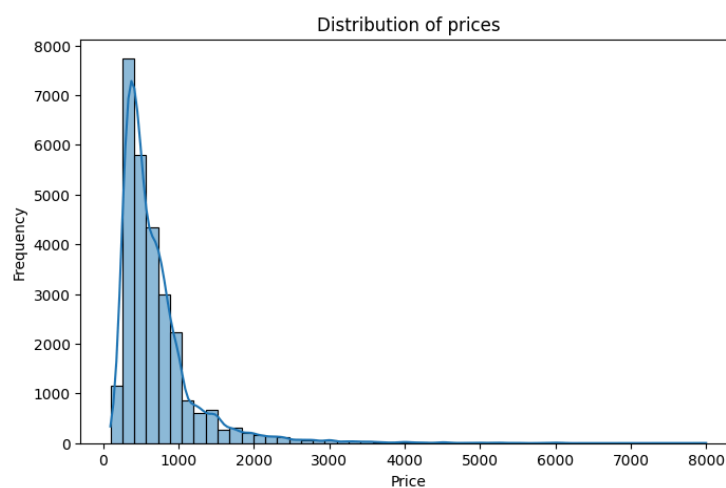
In this cloud we can also see which are the most used words, as we saw in previous plots.

		title	brand	average_rating	selling_price
5954	Women	Regular Fit Solid Button Down Collar For...	MILD	5	563.0
17399	Women	Regular Fit Solid Spread Collar Collar F...	Rose We	5	399.0
4220		Women Slim Fit Solid Casual Shirt	FOREVER YOU	5	725.0
17542		Women Solid Mid-Calf/Crew (Pack of 10)	Cots	5	475.0
23909		Printed Men Round Neck Red T-Shirt	Free Authori	5	519.0
23906		Printed Men Round Neck Yellow T-Shirt	Free Authori	5	454.0
11309		Full Sleeve Solid Women Jacket	ECKO Unl	5	1874.0
4354	Solid	Men Polo Neck Multicolor T-Shirt (Pack ...	Shoef	5	616.0
4357	Solid	Men Polo Neck Multicolor T-Shirt (Pack ...	Shoef	5	806.0
23852		Graphic Print Men Round Neck Blue T-Shirt	Free Authori	5	519.0

Here we see the products ordered by rating, in descending order. We see that there are many products with a 5 out of 5 in rating.



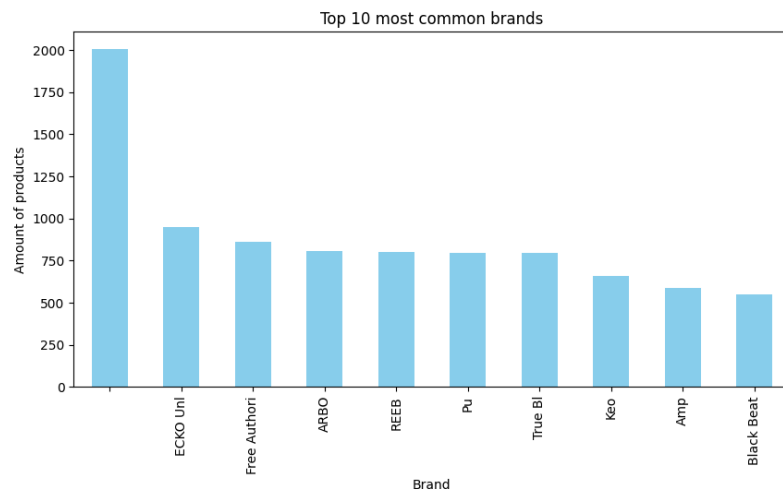
In this plot, we can deduce that there are only 4 categories, and they are very unbalanced because one of them is much more used.



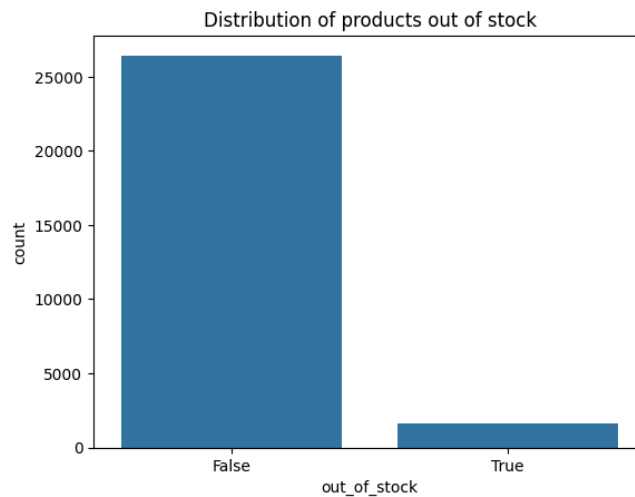
In the prices plot, we see that the cost is centered at 400, but there can be some products that have a price over 2000.

	title	brand	discount	selling_price
906	Printed Women Round Neck White T-Shirt	Jack Roy	87.0	128.0
903	Printed Women Round Neck White T-Shirt	Jack Roy	86.0	136.0
902	Printed Women Round Neck Grey T-Shirt	Jack Roy	86.0	136.0
18249	Printed Men Round Neck Multicolor T-Shirt (Pa...	yellowvib	85.0	449.0
9811	Solid Balclava Cap	Gracew	84.0	228.0
18016	Printed Women Round Neck Blue T-Shirt	yellowvib	84.0	280.0
18017	Printed Men Mandarin Collar Blue T-Shirt	yellowvib	84.0	282.0
18093	Printed Men Collared Neck Multicolor T-Shirt	yellowvib	84.0	230.0
91	nu-Lite Satin Tie & Cufflink (Red)		84.0	499.0
3102	Color Block Women Round Neck Green T-Shirt	REF	84.0	228.0

In this ranking we see the ordered products by discount in descending order, and we can see that there are products with very high discounts.



From this common brands plot, we deduce that there are many missing values that are counted as an empty brand.



In this out of stock plot, we see that the majority of products are available in stock, so it is a very unbalanced feature.

1.3. Tag of the repository

This report references the notebook in:

https://github.com/IsaacPerez02/SearchEngine-IRWA_2025/releases/tag/IRWA-2025-part-1